

LAB 9

Task # 01, 02, 03, 04 class

Main {

```
    public static void main(String[] args) {
        // You cannot create an object of Shape directly, as it's abstract
        // Shape shape = new Shape(); // This will cause a compilation error

        Shape[] arr = new Shape[3];
        arr[0] = new Circle();    arr[1]
        = new Rectangle();    arr[2] =
        new Circle();

        for (int i = 0; i < arr.length; i++) {
            arr[i].calculateArea();
            arr[i].details();
        }
    }
}
//abstract class abstract
class Shape{
    String color;
    int BorderThickness;

    // Abstract method - this method has no implementation
    abstract int calculateArea();
    abstract void details();
}
// Concrete subclass Circle that extends Shape
class Circle extends Shape{    int Radius = 2;
    @Override    int calculateArea() {    return
    Radius*Radius*3;
    }
    @Override
    void details() {
        System.out.println("Circle Details");
        System.out.println("Color: " + color);
        System.out.println("Border Thickness: " + BorderThickness);
    }
}
// Concrete subclass Circle that extends Shape class
Rectangle extends Shape{
```

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```
int length = 2;
int width = 2;

@Override int
calculateArea() {
    return length * width;
}
@Override
void details() {
    System.out.println("Rectangle Details");
    System.out.println("Color: " + color);
    System.out.println("Border Thickness: " + BorderThickness);
} }
```

Task # 05

```
// Abstract class Vehicle
abstract class Vehicle {

    // Abstract method to start the vehicle (no implementation)
    public abstract void start();

    // Abstract method to stop the vehicle (no implementation)
    public abstract void stop();
}

// Car class extends Vehicle class
Car extends Vehicle {

    // Implementing the start() method for Car
    @Override public
    void start() {
        System.out.println("The car is starting.");
    }

    // Implementing the stop() method for Car
    @Override
    public void stop() {
        System.out.println("The car is stopping.");
    }
}

// Bike class extends Vehicle class
Bike extends Vehicle { //
```

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Implementing the start() method

for Bike

```
@Override public
void start() {
    System.out.println("The bike is starting.");
}

// Implementing the stop() method for Bike
@Override
public void stop() {
    System.out.println("The bike is stopping.");
}
}
```

// Test class to create Car and Bike objects and call their methods public
class Main {

```
    public static void main(String[] args) {
        // Create a Car object
        Vehicle myCar = new Car();    // Call start
        and stop methods for Car    myCar.start(); //
        Output: The car is starting.    myCar.stop(); //
        Output: The car is stopping.

        // Create a Bike object
        Vehicle myBike = new Bike();    // Call start
        and stop methods for Bike    myBike.start(); //
        Output: The bike is starting.    myBike.stop(); //
        Output: The bike is stopping.
    }
}
```